

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech. II Semester I Year 2025-26
Tutorial Sheet-14 (M.M: 64)
UNIT-IV: Complex Analysis

CO4: The students will learn the complex variable – Differentiation.

PART – A

(2×5=10)

A1: Find the modulus and principal value of argument for $-1 + i\sqrt{3}$

Ans.: $2, \frac{2\pi}{\sqrt{3}}$

A2: If $u = 3x - 2xy$, $v = x^2 - y^2 + 3y + c$ then find $f(z)$ in terms

Ans.: $iz^2 + 3z + ic$

A3: Write C-R equations in polar form.

A4: Define complex exponential function.

A5: Discuss nature of the transformation $w = \frac{3z-4}{z-1}$

Ans.: Fixed point-2, Parabolic $w = \frac{1}{z-2} + 1$

PART – B

(7×3=21)

B1: Show that the function $u(x, y) = 4xy - 3x + 2$ is harmonic. Construct corresponding analytic function $f(z) = u(x, y) + iv(x, y)$: Express $f(z)$ in terms of complex variable z .

Ans.: $-2iz^2 - 3z + 2 + ic$

B2: Show that $\cos z = \frac{e^{iz} + e^{-iz}}{2}$ and $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$

B3: If $u = 2x^2 + y^2$ and $v = \frac{y^2}{x}$, show that curves $u = \text{constant}$ and $v = \text{constant}$ cut orthogonally at all

intersections but that the transformation $w = u + iv$ is not conformal.

PART – C

(11×3=33)

C1: Show that the function $f(z) = u + iv$, where

$$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}, \quad z \neq 0$$

$$= 0, \quad z = 0$$

Satisfies the C-R equations at $z = 0$. Is the function analytic at $z = 0$? Justify your answer.

C2: If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and $u + v = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$ find $f(z)$ in terms of z .

Ans.: $\frac{1}{2}(1+i) \cot z + \frac{c}{1+i}$

C3: Find the bilinear transformation which maps the points $z = 1, i, -1$ into the points $z = i, 0, -i$.

Hence find the image of $|z| < 1$

Ans.: $w = \frac{iz+1}{-iz+1}$